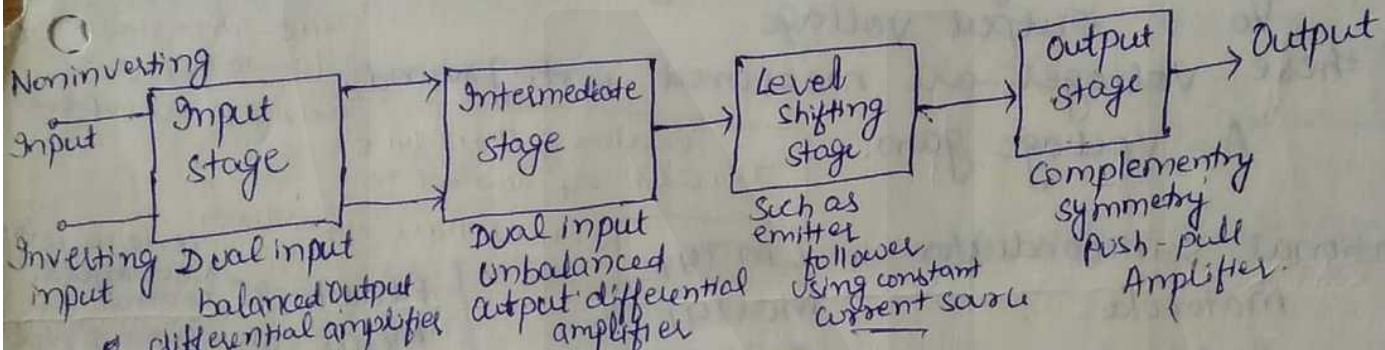
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Operational Amplifier :- is a direct coupled, high gain usually consisting of one or more differential amplifier and followed by a level translator and an output stage. It can be used to amplify dc as well as ac input signals. Operational amplifier is named such as because it was originally designed for mathematical operation such as addition, subtraction, integration and differentiation. Now a days it can be used for dc and ac signal amplification, active filters, oscillators, comparators, regulators etc.



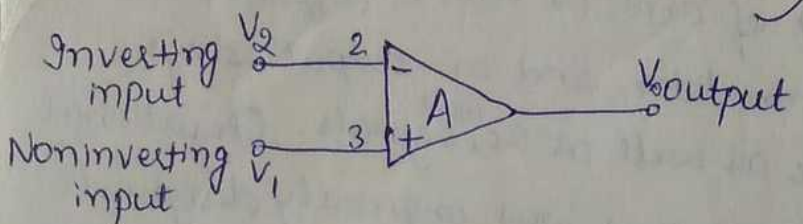
Block Diagram of an op-amp.

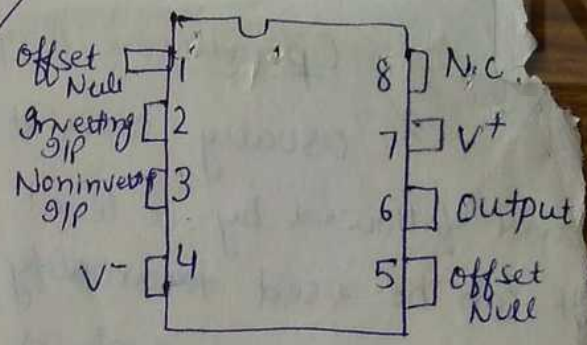
The first stage provide most of the voltage gain (60dB) of the amplifier. & it provides the ~~most of voltage~~ input resistance of op amp. The intermediate stage is usually driven by output of first stage. Next stage is level shifting stage. As direct coupling is used, the dc voltage at output of intermediate stage is well above the ground potential. Hence a level shift stage is used after the intermediate stage to shift the dc level at O/P of intermediate stage ~~to shift the dc level at output of inter~~ downward to zero volts wrt. ground. The output stage is usually a complementry amplifier output stage. The output stage increases the output voltage swing & raising the current supplying capability of op amp. It also establishes the low output resistance. Output of opamp is measured at output of this final stage push pull amplifier wrt. ground.

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Symbol of Op-amp :-





The ac signal (dc voltage) applied to (+) input produces an input (same polarity) signal at output. On the other hand (−) input produces an 180° out of phase (opposite polarity) signal at o/p.

V_1 = voltage at noninverting input
V_2 = voltage at inverting input
V_o = output voltage.
All these voltages are measured wrt. ground.
A = Voltage gain.

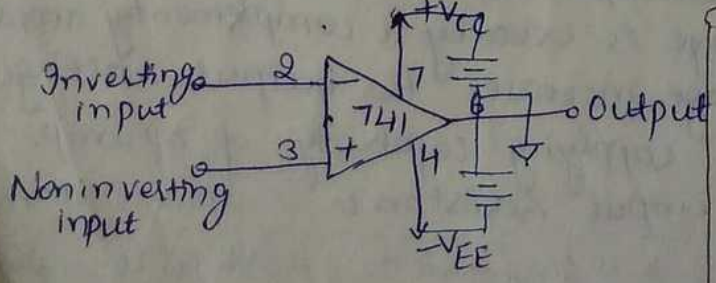
A, C, E, S & SC
741 - Military grade op. (−55° to 125°C)
741C - Comm. grade op. (0°C to 70/75°C)
741 A & E are improved versions of 7412 741 C
741C & E are identical to 741 S & 741SC are military

National semiconductor	LM741
Motorola	MC1741
RCA	CA3741
Texas instruments	SN52747
Signetics	N5741

Flat Pack - (8, 10, 14, 16 pins) rectangular ceramic case
Metal can - Metal case 3, 5, 8, 10, 12
Dual in line Package (DIP) mounted inside a plastic or ceramic case. It is mostly used ∵ it can be mounted easily. 8 Pin DIP. 12, 14, 16, 20.

Power Supplies Connections :-

For operation of op amp bipolar voltages are required i.e. two voltages are required, one +ve & other equal in magnitude but negative. These voltages must be referenced to a common point of ground. Fairchild :- (V+, V−), Motorola uses ($+V_{CC}$ & $-V_{EE}$).

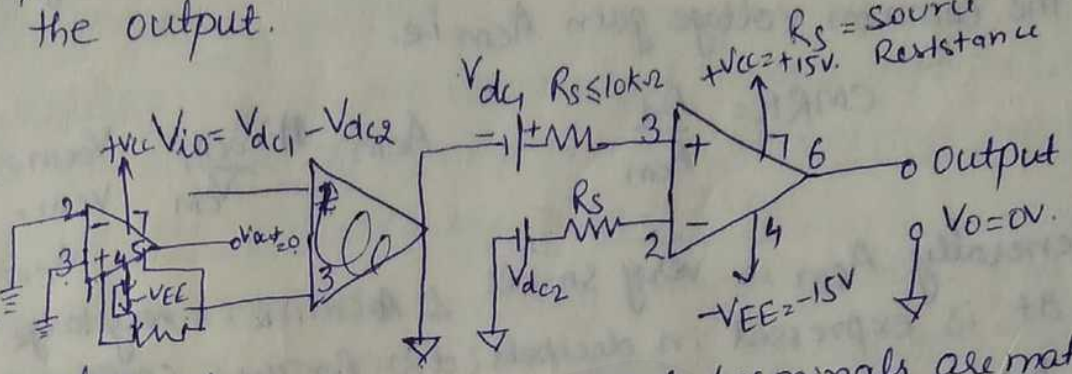


Temp. Range
Military temp. −55 to 125°C
Industrial −20 to 85°C
Comm. temp. Range − 0 to 70°C

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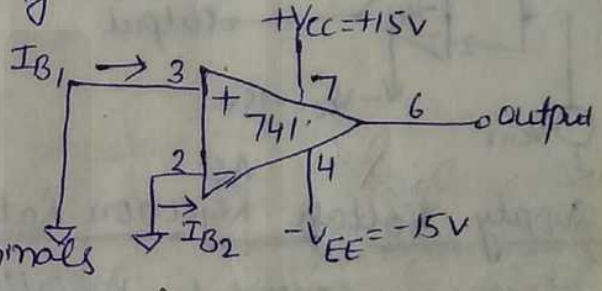
(1) DC gives Input offset Voltage (V_{io}) :- It is the voltage that must be applied between two input terminal of an op amp to null the output. (2)



for 741C the max. value of $V_{io} = 6mV_{dc}$

Smaller the value of V_{io} better the input terminals are matched.

(3) DC Input offset Current (bias current) :- The algebraic difference between the currents into inverting and noninverting terminals is referred as input offset current I_{io}.



$$I_{io} = I_{B1} - I_{B2}$$

Smaller the I_{io} better the input terminals are matched. for 741C is 200nA max. 6uA

(4) DC Input Bias Current :- Input bias I_B is the average of currents that flow into inverting and noninverting input terminals of op amp.

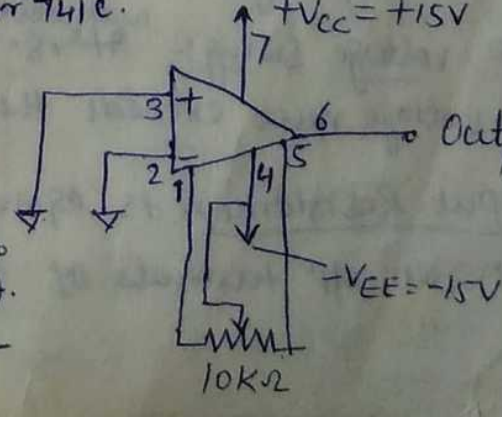
$$I_B = \frac{I_{B1} + I_{B2}}{2}$$

$I_B = 500\,nA$ I_{B1} & I_{B2} are base c/ts of first diff. amp. stage.

(5) DC Drifts: Thermal Drift:- off c/n & offset volt. change with temp. eg. we have nulled at 25°c. Later at 40°c ckt is no longer nulled. It is called thermal drift.

Differential Input Resistance :- It is the equiv resistance that can be measured at either inverting & noninverting input terminal with other terminal connected to ground. for 741C it is 2MΩ.

Input Capacitence (C_i) :- It is the equivalent capacitence that can be measured at either inverting & noninverting terminal with the other terminal connected to ground. 1.4pf for 741C.



Offset Voltage Adjustment Range :- is the range through which input voltage offset voltage can be adjusted by varying the 10KΩ potentiometer. output offset voltage can be reduced to zero volt. for 741C the offset volt. adjustment range is ±15mV.

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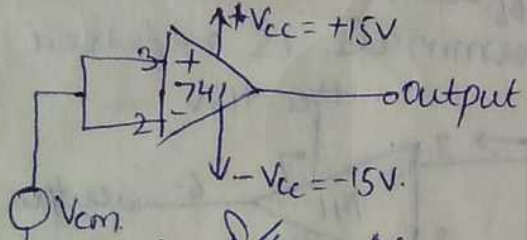
Common Mode Rejection Ratio :-

It is defined as the ratio of differential voltage gain A_d to the common voltage gain A_{cm} i.e.

$$CMRR = \frac{A_d}{A_{cm}} \qquad A_{cm} = \frac{V_{ocm}}{V_{cm}}$$

V_{ocm} = O/P C.M. voltage
V_{cm} = I/P " "
A_{cm} = Common mode voltage Gain

Generally A_{cm} is very small & CMRR is very large.
It is expressed in decibels (dB). For 741C CMRR is 90dB typically.
The higher the value of CMRR, the better is matching bet. two input terminals & smaller is the output common_mode voltage.



Supply Voltage Rejection Ratio :-

It is the change in input offset voltage V_{io} caused by variation in supply voltage is called supply voltage rejection ratio (SVRR). It is called power supply rejection ratio (PSRR) or power supply sensitivity (PSS).

$$SVRR = \frac{\Delta V_{io}}{\Delta V}$$

$$SVRR = 20 \log \left(\frac{\Delta V}{\mu V}\right)$$

Lower the value of SVRR in microvolts/volt, the better for opamp performance

Large Signal Voltage Gain :-

Opamp amplifies the difference bet. two i/p terminals.

A_o = Voltage Gain = $\frac{\text{output voltage}}{\text{differential i/p voltage}} = \frac{V_o}{V_{id}}$ $V_o = \pm 10V$, 741C $= 2,00,000$

∵ O/P signal amp. is much larger than i/p signal, the voltage gain is called large signal voltage gain.

O/P voltage Swing :-

It is indicated value of +ve & -ve sat. voltage of opamp. O/P voltage never exceeds these limits for given supply voltages $+V_{CC}$ & $-V_{EE}$
$\pm 13V$ for $R_L \geq 2K\Omega$

Output Resistance :-

is equivalent resistance that can be measured between O/P terminals of opamp & the ground. It is 75Ω for 741C opamp.

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Supply Current:- I_s is the current drawn by the op amp from the power supply. For 741C op-amp the supply current $I_s = 2.8$ mA.

Power Consumption:- P_c is the amount of quiescent power ($V_{in}=0V$) that must be consumed by op amp in order to operate properly. The amount of power consumed by 741C is 85mW.

AC:

Transient Response:- The response of any practically useful N/w to a given I/P is composed of two parts: transient & steady state response.

Transient response:- is that portion of complete response before O/P attains some fixed value. Once reached this fixed value remains at that level & is referred as steady state value. The response of N/w after it attains a fixed value is independent of time & is called steady state response. Transient response is time variant.

(2) Rise time & % of overshoot are char. of transient response. The time required by O/P to go from 10% to 90% of its final value is called rise time. Overshoot is max. amount by which O/P deviates from steady state value. It is expressed as a %. Rise time $\propto \frac{1}{\text{unity gain BW}}$. Smaller the rise time higher is BW.

AC

(3) Slew Rate:- It is the max. rate of change of O/P voltage per unit time & is expressed volts per microseconds.

$$SR = \left.\frac{dV_o}{dt}\right|_{max} \quad V/\mu s.$$

$V_{in} = V_{p\sin\omega t}$

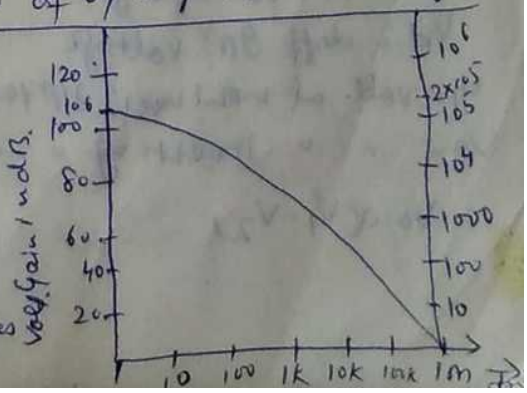
$V_{out} = V_{p\,out}\sin\omega t$

$\left.\frac{dv_{out}}{dt}\right|_{max} = V_{p\,out}\,\omega\cos\omega t$

$= V_{p\,out}\,\omega = 2\pi f V_{p\,out} = SR.$

Slew rate indicates how rapidly the output of op amp can change in response to change in the input frequency. Slew rate is one of the important factors in selecting the op amp for ac applications. 741C has low slew rate (0.5 V/μs) which limit its use at high freq. applications i.e. oscillators, filters, comparators etc. LM318 has slew rate 70 V/μs.

AC (1) Gain Bandwidth product:- GB product is the BW of op amp when voltage gain is 1. For an ideal op amp, open loop gain is very large & does not vary with freq. Not These assumptions are not good for an actual op amp. The dc & very low freq. open loop gain is abt. 2×10^5 which is large. With increase in freq. the gain decreases prop. 741C has a critical freq. f_c of 10Hz. Above f_c gain decreases 20 dB/decade until it reaches 1 at freq. of 1MHz.



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Unity gain B.W: unity is the freq. where voltage gain is unity. unity is 1MHz. 741C unity is 1MHz. So 741C can amplify the signal upto 1MHz. Beyond 1MHz voltage gain is less than unity & it is useless.

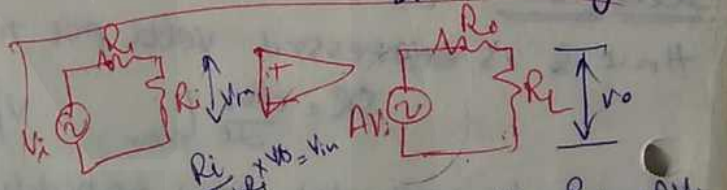
SR = $\left.\frac{dV_{out}}{dt}\right|_{max}$. $V_{in} = V_p \sin\omega t$ $V_{out} = V_p \sin\omega t$

$\frac{dV_{out}}{dt} = V_p \omega \cos\omega t$ $\left.\frac{dV_{out}}{dt}\right|_{max} = V_p \omega$ volt/sec $= 2\pi f V_p$

Ideal Op-amp:

$V_{id} = 0$

1. Open loop gain is ∞
2. Infinite i/p resistance means i/p current (current drawn from source) is zero. & so it does not load the source. Ideal diode is voltage controlled device so that almost any signal source can drive it.
3. Output resistance is zero so that o/p can drive an ∞ no. of other devices. V_{out} is independent of current drawn by the load.
4. Zero output voltage when i/p voltage is zero. (perfect balance).
5. Infinite BW so that any freq. signal from 0 to ∞ Hz can be amplified without attenuation (Dc and ac both signal can be amplified)
6. CMRR is ∞. So that amplifier is free from thermal noise, pick up etc.
7. Slew rate is ∞ so that output voltage change occurs simultaneously with i/p voltage changes.
8. Output voltage is zero when i/p voltage is zero i.e. offset voltage is zero.
9. Drift of char. with temp. is nil.



(1) Equivalent ckt of an op amp (Thevenin):-

AV_{id} is thevenin equivalent voltage source

R_o is thevenin equivalent resistance looking back into the o/p terminal of an op-amp.

$V_o = AV_{id} = A(V_1 - V_2)$

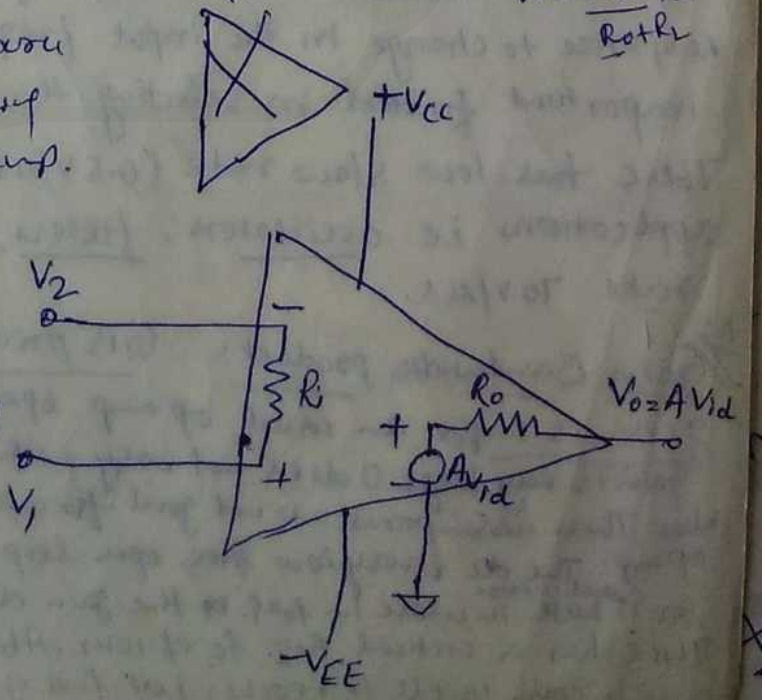
A = Voltage gain

V_{id} = diff. i/p voltage

V_1 = volt. of non inverting i/p term wrt ground

V_2 = " " inverting " " " "

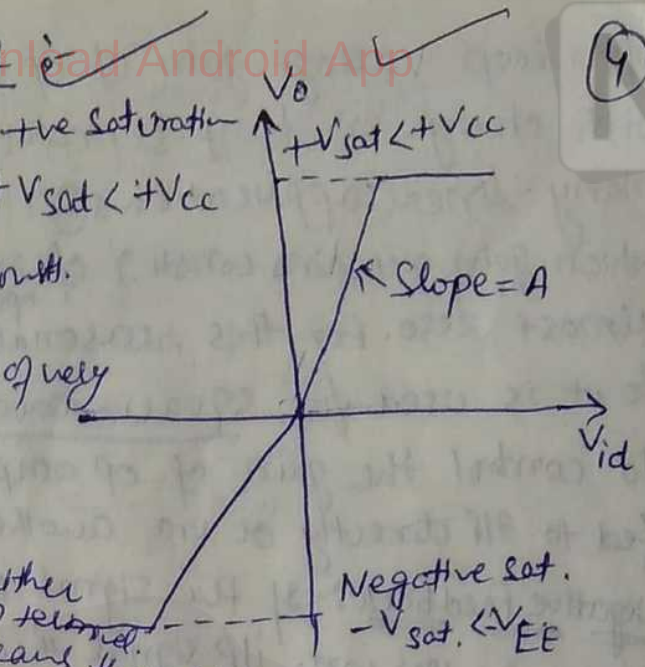
$V_o \propto V_1 - V_2$



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Ideal voltage Transfer Curve :-



s ideal voltage transfer curve

eal :- o/p offset voltage is zero. Gain constt.

curve will be almost vertical because of very large value of A.

Open loop indicates there is no connection either direct or via another N/w exists bet. o/p & i/p terminal. means there is no F/B

Openloop op amp Configurations :-

In open loop it is high gain amplifier.

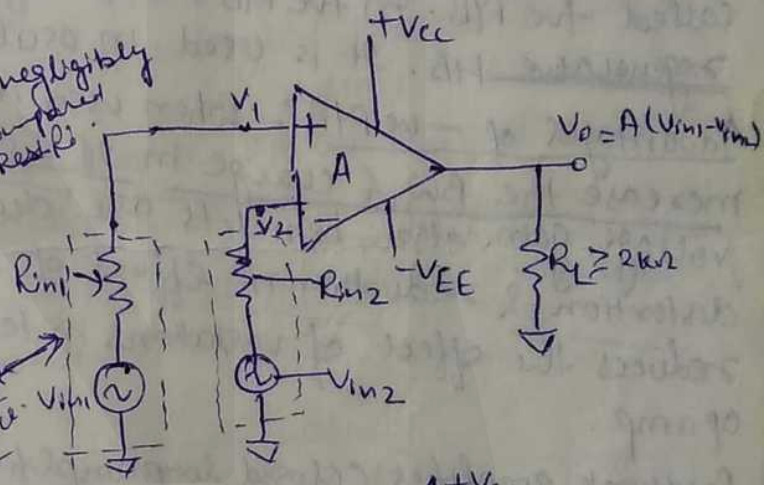
Three open loop config
- Differential amplifier
- Inverting "
- Non inverting amplifier.

Differential Amplifier :-

$V_1 = V_{in1}$ & $V_2 = V_{in2}$ R_{in1} & R_{in2} are negligibly small as compared to o/p Resistance

$V_{o2} = A(V_{in1} - V_{in2})$

$= A(V_1 - V_2)$



Polarity of o/p voltage is dependent on polarity of i/p diff. voltage (V_{in1} V_{in2} Signal source)

Gain A is referred as open loop gain.

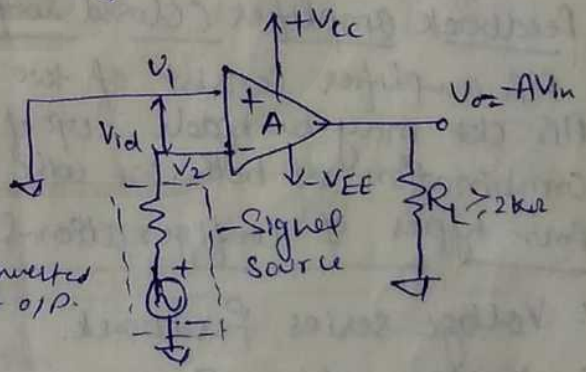
Inverting Amplifier :-

$V_1 = 0V$ & $V_2 = V_{in}$

$V_{o2} = -AV_{in}$

-ve sign indicates that o/p voltage is out of phase wrt. by 180° or is of opp. polarity

In inv. amp. i/p signal is amplified by gain A & is also inverted at o/p.

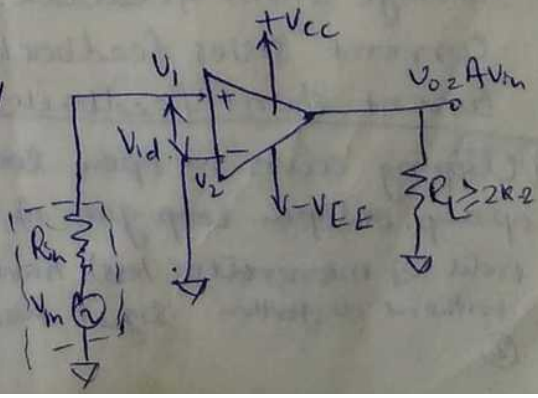


Non inverting Amplifier :-

$V_1 = V_{in}$ $V_2 = 0V$.

$V_{o2} = AV_{in}$

(1) In open loop config. any i/p signal that is only slightly greater than zero drives the o/p to sat. level. This is due to very high gain (A) of op amp. Thus, when operated open loop, the o/p of op amp is either -ve or +ve sat. or switches bet. +ve & -ve sat. levels. For this reason, open loop op-amp config are not used in linear applications.



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① Open loop voltage gain of op amp is not constant it varies with change in temp. & power supply. So it is unsuitable for many linear application & ③ The bandwidth (band of freqs. for which gain remains const.) of most open loop op amp is negligibly small almost zero. For this reason open loop op amp is impractical in ac applications. So it is used for square wave applications.

To control the gain of op amp we use the feedback i.e. an o/p signal is fed to i/p directly or via another N/w.

Negative feedback:- If the signal fed back is of opposite polarity or out of phase by 180° wrt. i/p signal then it is called -ve feedback (degenerative feedback); when it is used it degenerates (reduces) the o/p voltage amplitude & in turn reduces the voltage gain.

+ve feedback:- If the signal fed back is in phase with i/p signal F/B is called +ve F/B. In +ve F/B the F/B signal aids the i/p signal. also called regenerative F/B. It is used in oscillator circuits.

Advantages of -ve F/B:- When used in amplifiers, -ve F/B stablizes the gain, increase the BW & change in i/p & o/p resistances. but it reduces the voltage gain. other benefits are decrease in nonlinear & harmonic distortion & reduction in effect of i/p offset voltage at the o/p. It also reduces the effect of variations in temp. & supply voltage on the o/p of op amp.

Feedback amplifier (closed loop amplifier) :- An op F/B amplifier consist of two parts: an opamp & a feedback circuit. F/B ckt may be made up of either passive or active components or combination of both. We will use only purely resistive F/B ckt.

Four types of configurations:-

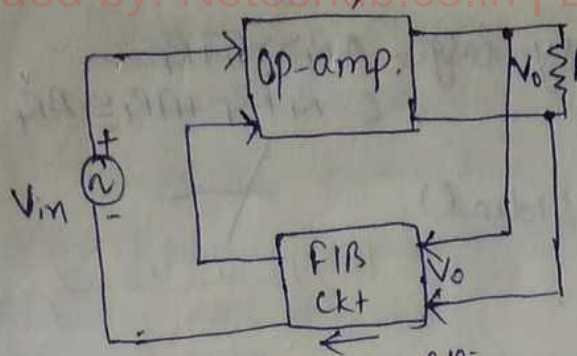
1. Voltage series feedback.
2. Voltage shunt feedback.
3. Current series feedback.
4. Current shunt feedback.

① Clipping occurs in open loop confs. when o/p exceeds the sat. level of op amp. & open loop gain of opamp is very high (only smaller signal (of order of microvolt or less) having very low freq. may be amplified accurately without distortion. Signal thus small are very susceptible

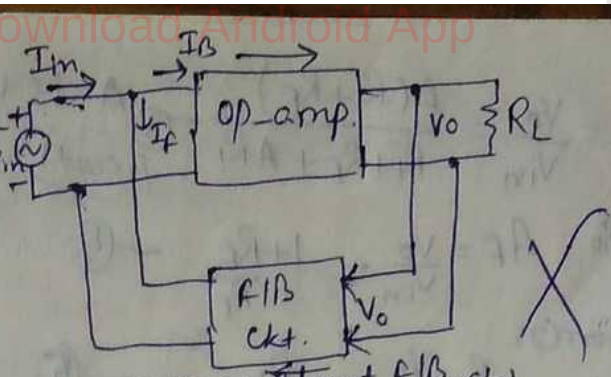
②

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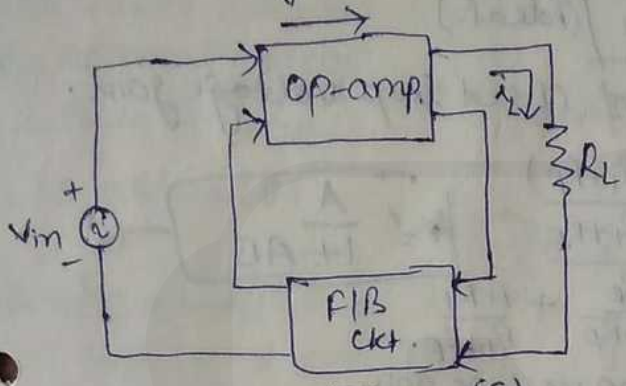
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Voltage series FIB (a)

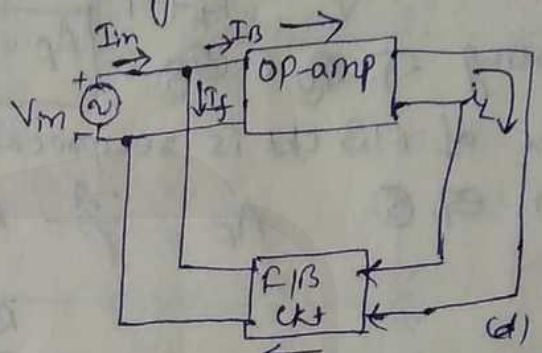


Voltage Shunt FIB. (b)



(c)

Current Series FIB

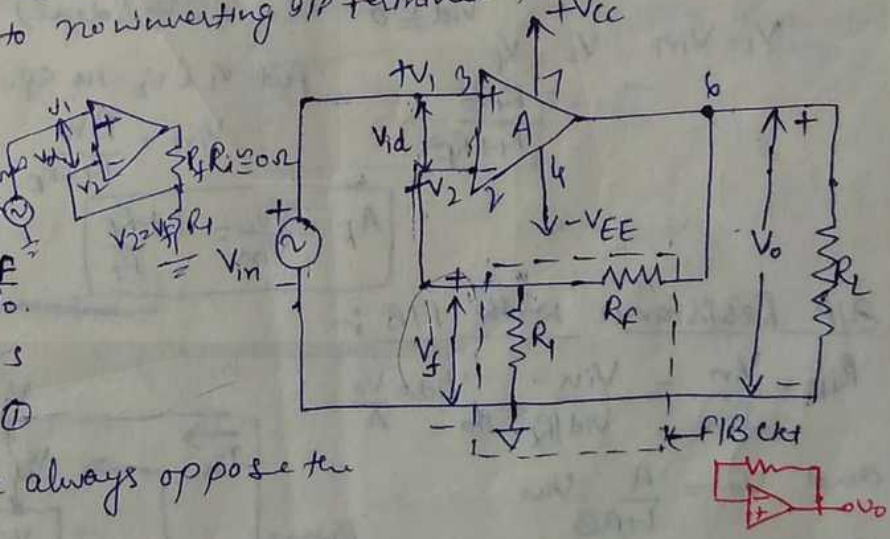


(d)

Current Shunt FIB.

In fig. (a) & (b) voltage across R_L is i/p voltage to FIB ckt. The FIB qty (volt. or cu) is o/p of FIB ckt & is pro. to o/p voltage.
In fig. (c) & (d) the load c/u i_L flows into the FIB ckt. The o/p of FIB ckt. (volt. of cu) is pro. to load current i_L.

Non inverting Amplifier :- (Closed loop non inverting amplifier) :- It uses the FIB & i/p signal is applied to non inverting i/p terminal of op-amp.



Open loop voltage gain $A = \frac{V_o}{V_{id}}$

Closed " " " $A_F = \frac{V_o}{V_{in}}$

Gain of FIB circuit $B = \frac{V_f}{V_o}$

Apply KVL for i/p loop is

$$V_{id} = V_{in} - V_f \quad \text{— (1)}$$

In above eqn. FIB voltage always oppose the i/p volt. hence FIB is -ve.

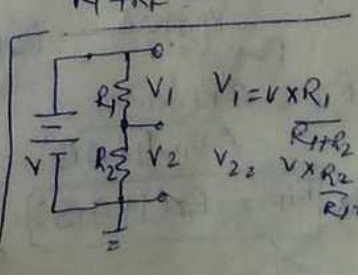
We know $V_o = A(V_1 - V_2)$ $V_1 = V_{in}$ $V_2 = V_f = \frac{R_1 V_o}{R_1 + R_f}$ — (2)

$$V_o = A\left(V_{in} - \frac{R_1 V_o}{R_1 + R_f}\right)$$

$$V_o = A\left(\frac{V_{in}(R_1 + R_f) - R_1 V_o}{R_1 + R_f}\right)$$

$$V_o\left(1 + \frac{AR_1}{R_1 + R_f}\right) = A V_{in}$$

$$V_o\left(\frac{R_1 + R_f + AR_1}{R_1 + R_f}\right) = A V_{in}$$



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$A_f = \frac{V_o}{V_{in}} = \frac{A(R_1+R_f)}{R_1+R_f+AR_1}$ — (3) (exact)

A is very large $AR_1 >> R_1+R_f$ & $R_1+R_f+AR_1 \cong AR_1$

So $A_f = \frac{V_o}{V_{in}} = 1+\frac{R_f}{R_1}$ — (4) (ideal).

17, 18, 40, 46, 51, 54

From (2)

$B = \frac{V_f}{V_o} = \frac{R_1}{R_1+R_f}$ — (5)

Comparing eq. (4) & (5) $\boxed{A_f = \frac{1}{B}}$ (ideal.)

So Gain of F/B ckt is reciprocal of closed loop voltage gain.

From eq. (3) $A_f = \frac{A(R_1+R_f)}{R_1+R_f} \Big/ \frac{R_1+R_f}{R_1+R_f} + \frac{AR_1}{R_1+R_f}$

$\boxed{A_f \cong \frac{A}{1+AB}}$ — (6)

$\boxed{A_f = \frac{A}{1+AB}}$ — (6)

A_f = Closed loop voltage gain
A = Open " "
B = Gain of F/B ckt.
AB = loop gain.

Difference input voltage ideally zero :-

$V_{id} = \frac{V_o}{A}$ Since A is very large (ideally ∞)

$V_{id} \cong 0$ i.e. $V_1 \cong V_2$ (ideal) — (a)

$V_1 = V_{in}$ $V_2 = V_f = \frac{R_1 V_o}{R_1+R_f}$

Put V_1 & V_2 in eq. (a)

$V_{in} = \frac{R_1 V_o}{R_1+R_f}$

$\boxed{A_f = \frac{V_o}{V_{in}} = 1+\frac{R_f}{R_1}}$

O/P Resistance with F/B :-

$R_{if} = \frac{V_{in}}{I_{in}} = \frac{V_{in}}{V_{id}/R_i}$ — (b) $V_{id} = \frac{V_o}{A}$

and $V_o = \frac{A}{1+AB} V_{in}$.

Put V_{id} in eq. (b)

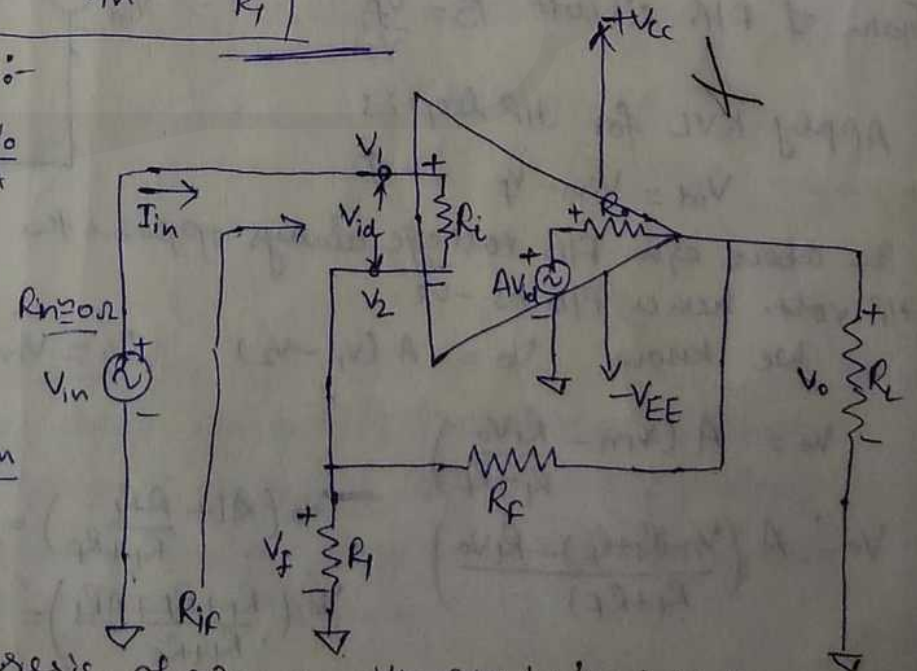
$R_{if} = R_i \frac{V_{in}}{V_o/A}$

Put V_o.

$R_{if} = \frac{AR_i V_{in}}{AV_{in}/1+AB}$

$R_{if} = AR_i \frac{V_{in}}{V_o}$

$\boxed{R_{if} = R_i(1+AB)}$ So O/P resis. of opamp with F/B is (1+AB) times that without F/B.



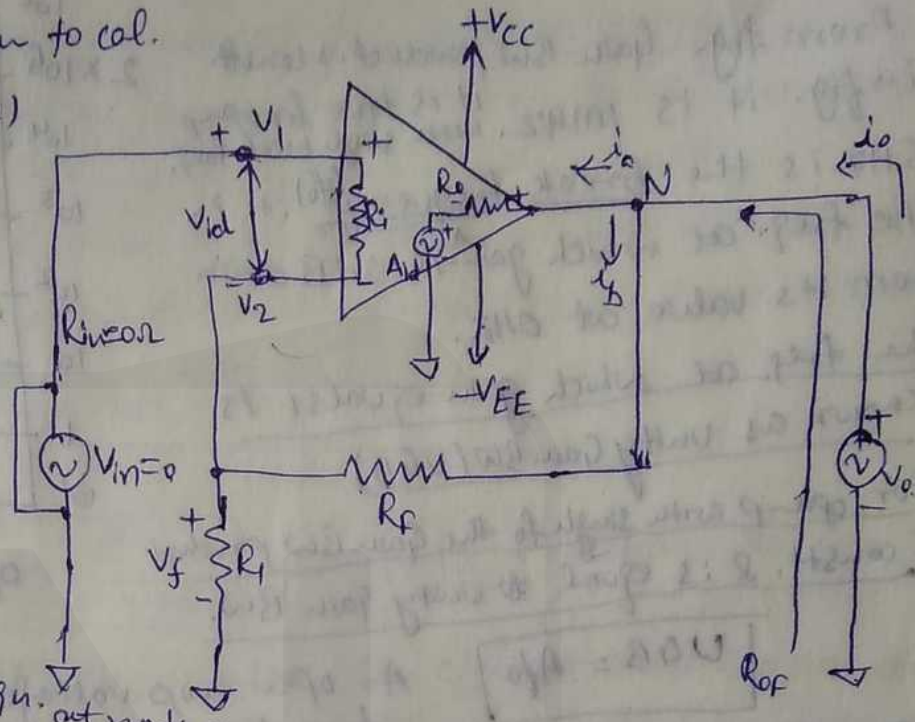
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Output Resistance with feedback :-

It is the resistance determined look back into the F/B amplifier from the o/p terminal.

We apply Thevenin Theorem to cal. o/p resistance with F/B. (R_{of}) for dependent source. (independent source) Reduce V_{in} to zero & apply external voltage v_o & then cal. i_o.



R_{of} is defined as

$$R_{of} = \frac{v_o}{i_o} \quad \text{—①}$$

Write Kirchoff's c/u eqn. at node N.

$$i_o = i_a + i_b.$$

Since $(R_f + R_1) \| R_i >> R_o$ & $i_a >> i_b$

$\therefore i_o \simeq i_a$

Current i_o can be found by writting Kirchoff's voltage eqn. from o/p loop.

$$v_o - R_o i_o - A V_{id} = 0$$

$$\boxed{i_o = \frac{v_o - A V_{id}}{R_o}} \quad \text{—②}$$

$$V_{id} = V_1 - V_2$$

$$V_{id} = 0 - \frac{R_1 V_o}{R_1 + R_f} = -B V_o.$$

Put V_{id} in eqn. ②

$$\therefore i_o = \frac{v_o + A B V_o}{R_o}$$

Put value in eq. ①

$$R_{of} = \frac{v_o}{(v_o + A B V_o)/R_o} = R_{of} = \frac{R_o}{1 + AB}$$

So o/p resistance of voltage series F/B amp. is $\frac{1}{1+AB}$ times the o/p resistance R_o of the op amp. i.e. o/p resistance of op amp with F/B is much smaller than o/p resistance without F/B.

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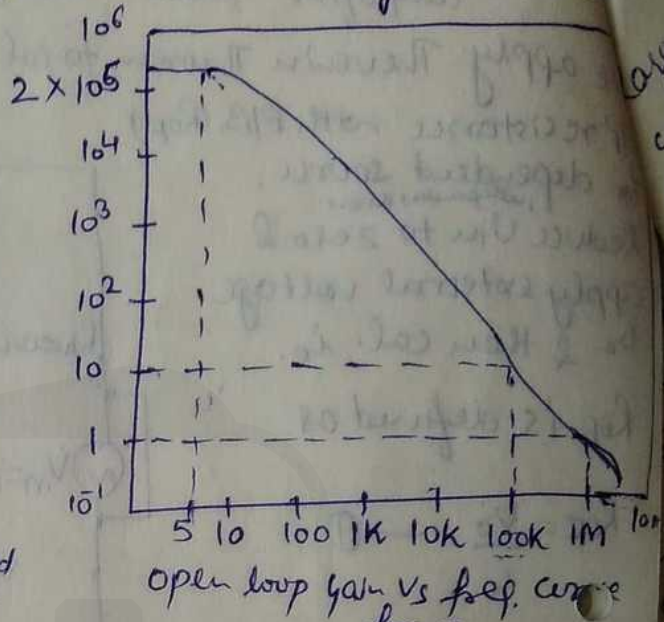
Bandwidth with Feedback :-

BW is defined as the band (range) of freqs. for which Gain remains constt.

From fig. Gain BW product is constt. in fig. it is 1MHz. it is true for opamp with single break freq. 5Hz is the break frequency (f_0); it is the freq. at which gain is 3dB down from its value at 0Hz.

The freq. at which gain equals 1 is known as unity Gain BW (UGB)

For opamp with single f_0 the Gain BW product is constt. & is equal to unity gain BW.



open loop gain Vs freq. curve for 741C.

$$UGB = Af_0$$

A = open loop voltage gain
f_0 = break freq. of an op amp.

Or only for a single break frequency op amp.

$$UGB = (A_F f_F)$$

A_F = Closed loop voltage gain
f_F = BW with F/B.

$$Af_0 = A_F f_F$$

$$f_F = \frac{Af_0}{A_F}$$

For non inverting amplifier with F/B. $A_F = \frac{A}{1+AB}$

Substitute A_F in above eqn.

$$f_F = \frac{Af_0}{A/1+AB} = \boxed{f_F = f_0(1+AB)}$$

eqn. show that BW of noninverting amplifier with feedback f_F is equal to (1+AB) times to its BW without F/B (f_0).

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Total Output offset voltage with FIB :-

In an op-amp when i/p is zero, o/p is also expected to zero. ∵ of effect of i/p offset voltage & c/n, the o/p is significantly larger ∵ of very high open loop gain. This enhanced o/p voltage is called total o/p offset voltage V_{OOT}. In open loop op-amp total o/p offset voltage is either equal to +ve or -ve sat. voltage. Due to FIB gain of non-inver. amp. changes from A to A/(1+AB). So total o/p offset voltage with FIB must also be 1/(1+AB) times the volt. without FIB.

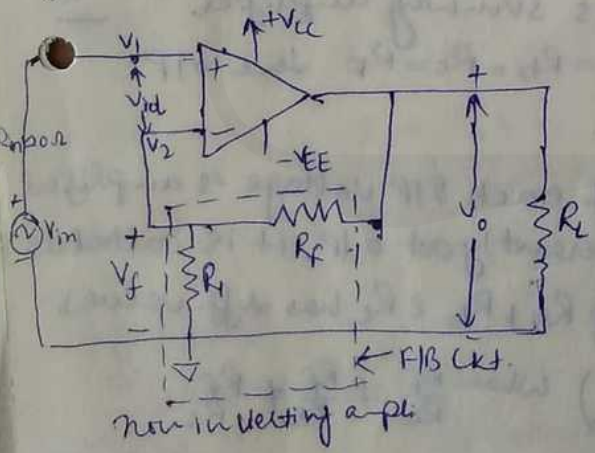
$$\text{Total o/p offset voltage with FIB} = \frac{\text{Total o/p offset voltage without FIB}}{1+AB}$$

$$\boxed{V_{OOT} = \frac{\pm V_{sat}}{1+AB}}$$

1/1+AB is always less than 1.

Higher the value of (1+AB), smaller the effect of noise & of variations in supply voltages & change in temp. on o/p voltage of a noninverting amplifier.

Voltage follower :- when the noninverting amplifier is configured for unity gain is called a voltage follower. ∵ o/p voltage is equal to & in phase with i/p. In voltage follower the o/p follows the i/p. It is similar to discrete emitter follower. The voltage follower is preferred ∵ it has much higher i/p resi. & o/p amplitude is exactly to i/p.



non inverting ampli. volt. fol.

open R_1 & short R_f

$B = 1$

$$A_f = \frac{A}{1+AB} = A_F = \frac{1}{\frac{1}{A}+B}$$

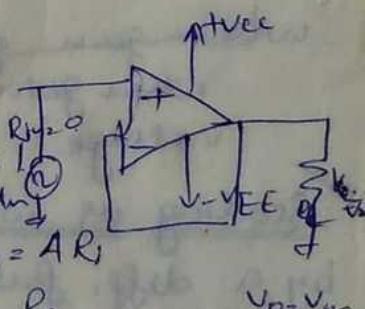
$$R_{iF} = R_i(1+AB) \qquad R_{iF} = AR_i$$

$$R_{oF} = \frac{R_o}{1+AB} \qquad R_{oF} = \frac{R_o}{A}$$

$$f_F = (1+AB)f_o \qquad f_F = Af_o$$

$$V_{OOT} = \frac{\pm V_{sat}}{1+AB} \qquad V_{OOT} = \frac{\pm V_{sat}}{A}$$

Since $(1+A) \simeq A$



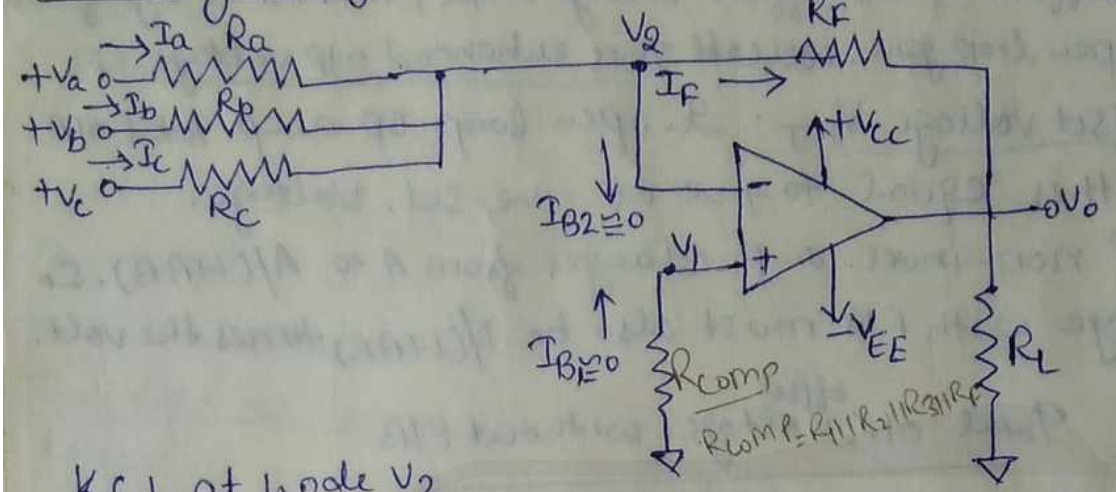
It is also called noninverting buffer ∵ when placed between two N/ws it removes the loading on first N/w.

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Summing, Scaling & Averaging Amplifier :-

Inverting Configuration :-



KCL at node V_2

$$I_a + I_b + I_c = I_B + I_f \quad \therefore \frac{V_a - V_2}{R_a} + \frac{V_b - V_2}{R_b} + \frac{V_c - V_2}{R_c} = \frac{V_2 - V_o}{R_f}$$

since R_i & A are ideally ∞, $I_B = 0A$, & $V_1 = V_2 \cong 0V$

$$\frac{V_a}{R_a} + \frac{V_b}{R_b} + \frac{V_c}{R_c} = \frac{-V_o}{R_f}$$

$$V_o = -\left(\frac{R_f V_a}{R_a} + \frac{R_f}{R_b} V_b + \frac{R_f}{R_c} V_c\right)$$

[Gain of inverting : $A_d = -\frac{R_f}{R_1}$

Summing amplifier :- If $R_a = R_b = R_c = R$

then $V_o = -(V_a + V_b + V_c)\frac{R_f}{R}$

So o/p voltage is -ve sum of all i/ps times the gain of ckt $\frac{R_f}{R}$. ~~when gain of ckt is 1~~ hence it is summing amplifier.

when gain of ckt is 1. i.e. $R_a = R_b = R_c = R_f$ the o/p voltage $V_o = -(V_a + V_b + V_c)$.

Scaling or weighted Amplifier :- If each i/p voltage is amplified by a diff. factor or weighted differently at o/p, it is called as Scaling or weighted amplifier. If R_a, R_b & R_c has diff. values.

$$V_o = -\left(\frac{R_f}{R_a} V_a + \frac{R_f}{R_b} V_b + \frac{R_f}{R_c} V_c\right) \text{ when } \frac{R_f}{R_a} \neq \frac{R_f}{R_b} \neq \frac{R_f}{R_c}.$$

Average circuit :- in which o/p volt. is equal to avg. of all i/p voltages. So $R_a = R_b = R_c = R$ or gain by which each i/p is amplified must be equal to 1 over no. of i/ps i.e.

$\frac{R_f}{R} = \frac{1}{n}$ where n = no. of i/ps. $\frac{R_f}{R} = \frac{1}{3}$

$V_o = -\left(\frac{V_a + V_b + V_c}{3}\right)$

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In ckt we want $\frac{R_f}{R} = \frac{1}{3}$ So

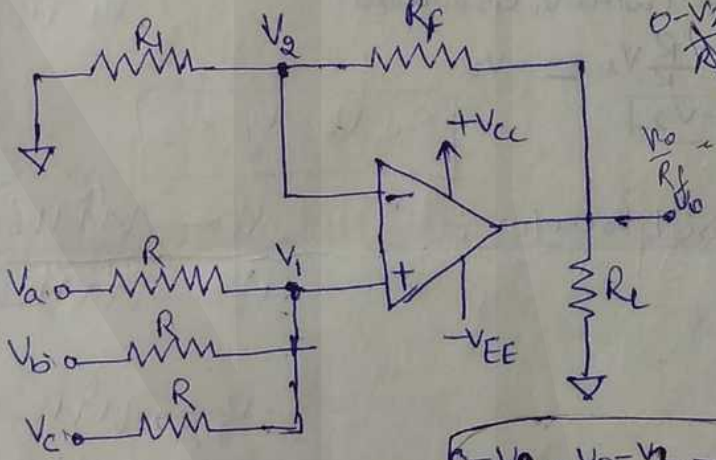
So $V_o = -\left(\frac{V_a + V_b + V_c}{3}\right)$

Use:- These ckts are used in analog computers & audio mixers in which a no. of I/Ps are added up (mixed) up to produce a desired O/P.

In practical ckt I/P bias ckt compensating resistor R_{comp} should be provided. To find R_{comp} make all I/Ps $V_1 = V_2 = V_3 = 0$ so that effective I/P resis. $R_i = R_1 \| R_2 \| R_3$ ∴ $R_{comp} = R_i \| R_f = R_1 \| R_2 \| R_3 \| R_f$.

Non Inverting Summing Amplifier:-

A summer that gives non inverted sum is non inverting summing amplifier



$\frac{V_a - V_1}{R} + \frac{V_b - V_1}{R} + \frac{V_c - V_1}{R} = 0$

So $V_1 = \frac{V_a + V_b + V_c}{3}$

Hence O/P voltage is $V_o = \left(1 + \frac{R_f}{R_1}\right) V_1$

$V_o = \left(1 + \frac{R_f}{R_1}\right)\left(\frac{V_a + V_b + V_c}{3}\right)$

Avg: Amplifier:-

O/P voltage is equal to avg. of all I/P time gain of ckt $\left(1 + \frac{R_f}{R_1}\right)$ If the gain is 1, O/P voltage is equal to avg. of all I/P voltage.

Summing amplifier:- if $\left(1 + \frac{R_f}{R_1}\right)$ is equal to no. of I/Ps the O/P voltage becomes equal to sum of all I/P voltage. i.e. if $\left(1 + \frac{R_f}{R_1}\right) = 3$

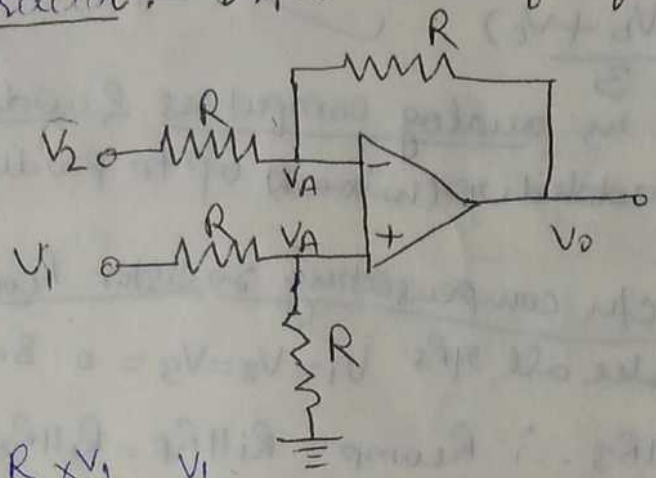
So $V_o = V_a + V_b + V_c$

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~~Voltage Shunt~~ (Differential amplifier)

Subtractor :- All R are of equal value



Apply Nodal

$$\frac{V_1 - V_A}{R} + \frac{0 - V_A}{R} = 0$$

$$\boxed{V_A = \frac{V_1}{2}}$$

$V_A = \frac{R}{R+R} \times V_1 \Rightarrow \frac{V_1}{2}$

V_1 alone make $V_2 = 0$

$V_{o1} = (1 + \frac{R}{R})\frac{V_1}{2} = V_1$

V_2 alone (with V_1 Grounded)

$V_{o2} = -\frac{R}{R} V_2 = -V_2$

$\boxed{V_o = V_1 - V_2}$

$$\frac{V_1 - V_A}{R} + \frac{V_2 - V_A}{R} + \frac{V_o - V_A}{R} = 0 \checkmark$$

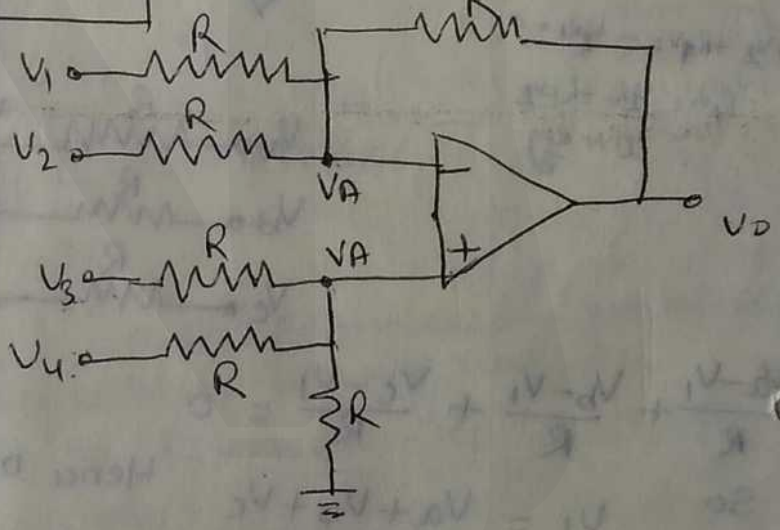
$V_2 + V_o - 2V_A = 0$

$V_o = 2V_A + V_2$ Put V_A

$= 2 \times \frac{V_1}{2} - V_2$

$\boxed{\text{So } V_o = V_1 - V_2}$

Adding Subtractor :-



$$\frac{V_1 - V_A}{R} + \frac{V_2 - V_A}{R} + \frac{V_o - V_A}{R} = 0$$

$V_1 + V_2 + V_o - 3V_A = 0$

$$\boxed{V_1 + V_2 + V_o = 3V_A} \quad -(1)$$

$$\frac{V_3 - V_A}{R} + \frac{V_4 - V_A}{R} + \frac{0 - V_A}{R} = 0 \qquad \boxed{V_3 + V_4 = 3V_A} \quad -(2)$$

from (1) & (2)

$V_1 + V_2 + V_o = V_3 + V_4$

$$\boxed{V_o = (V_3 + V_4) - (V_1 + V_2)}$$

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